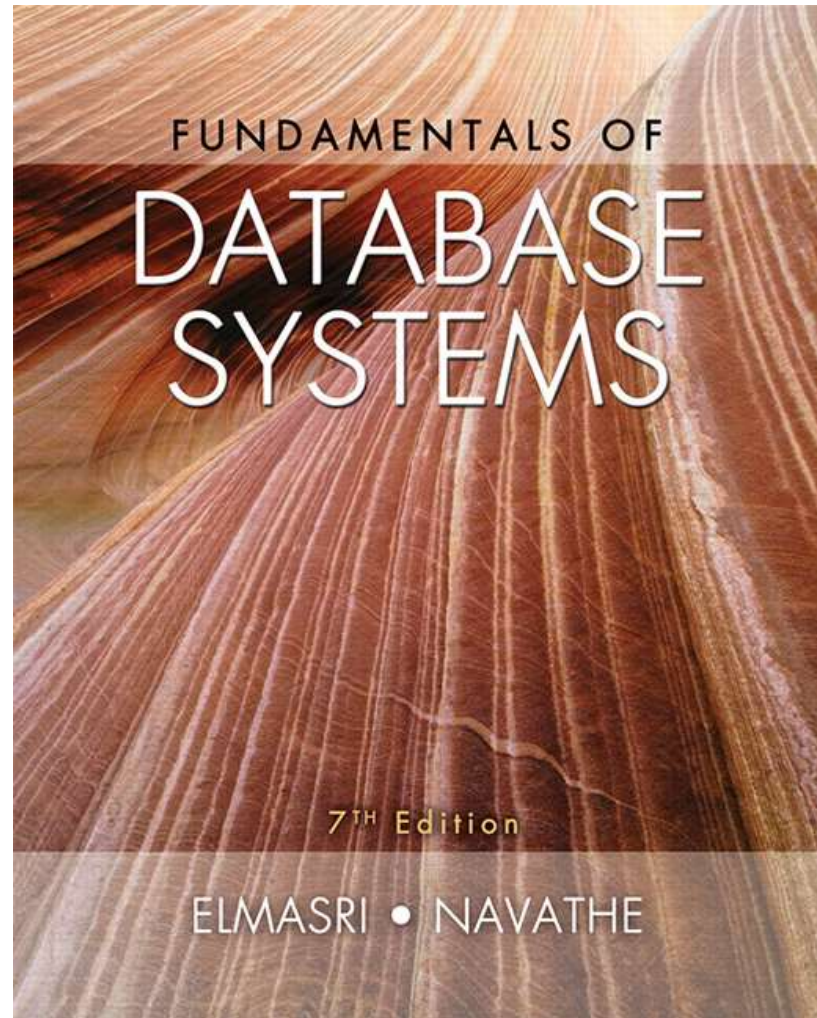




CHAPTER 7

More SQL: Complex Queries, Triggers, Views, and Schema Modification

Chapter 7 Outline

- More Complex SQL Retrieval Queries
- Specifying Semantic Constraints as Assertions and Actions as Triggers
- Views (Virtual Tables) in SQL
- Schema Modification in SQL

More Complex SQL Retrieval Queries

- Additional features allow users to specify more complex retrievals from database:
 - Nested queries, joined tables, and outer joins (in the FROM clause), aggregate functions, and grouping

Comparisons Involving NULL and Three-Valued Logic

- Meanings of NULL
 - Unknown value
 - Unavailable or withheld value
 - Not applicable attribute
- Each individual NULL value considered to be different from every other NULL value
- SQL uses a three-valued logic:
 - TRUE, FALSE, and UNKNOWN (like Maybe)
- **NULL = NULL comparison is avoided**

Comparisons Involving NULL and Three-Valued Logic (cont'd.)

Table 7.1 Logical Connectives in Three-Valued Logic

(a)	AND	TRUE	FALSE	UNKNOWN
	TRUE	TRUE	FALSE	UNKNOWN
	FALSE	FALSE	FALSE	FALSE
	UNKNOWN	UNKNOWN	FALSE	UNKNOWN
(b)	OR	TRUE	FALSE	UNKNOWN
	TRUE	TRUE	TRUE	TRUE
	FALSE	TRUE	FALSE	UNKNOWN
	UNKNOWN	TRUE	UNKNOWN	UNKNOWN
(c)	NOT			
	TRUE	FALSE		
	FALSE	TRUE		
	UNKNOWN	UNKNOWN		

Comparisons Involving NULL and Three-Valued Logic (cont'd.)

- SQL allows queries that check whether an attribute value is NULL
 - IS or IS NOT NULL

Query 18. Retrieve the names of all employees who do not have supervisors.

```
Q18:  SELECT  Fname, Lname
      FROM    EMPLOYEE
      WHERE   Super_ssn IS NULL;
```

Nested Queries, Tuples, and Set/Multiset Comparisons

- **Nested queries**

- Complete select-from-where blocks within WHERE clause of another query
- **Outer query and nested subqueries**

- **Comparison operator** `IN`

- Compares value v with a set (or multiset) of values V
- Evaluates to `TRUE` if v is one of the elements in V

Nested Queries (cont'd.)

```
Q4A:  SELECT DISTINCT Pnumber
      FROM PROJECT
      WHERE Pnumber IN
        ( SELECT Pnumber
          FROM PROJECT, DEPARTMENT, EMPLOYEE
          WHERE Dnum=Dnumber AND
                Mgr_ssn=Ssn AND Lname='Smith' )
      OR
      Pnumber IN
        ( SELECT Pno
          FROM WORKS_ON, EMPLOYEE
          WHERE Essn=Ssn AND Lname='Smith' );
```

Nested Queries (cont'd.)

- Use tuples of values in comparisons
 - Place them within parentheses

```
SELECT    DISTINCT Essn
FROM      WORKS_ON
WHERE     (Pno, Hours) IN ( SELECT    Pno, Hours
                           FROM      WORKS_ON
                           WHERE     Essn='123456789' );
```

Nested Queries (cont'd.)

- Use other comparison operators to compare a single value v
 - $=$ ANY (or $=$ SOME) operator
 - Returns TRUE if the value v is equal to some value in the set V and is hence equivalent to IN
 - Other operators that can be combined with ANY (or SOME): $>$, $>=$, $<$, $<=$, and $<>$
 - ALL: value must exceed all values from nested query

```
SELECT  Lname, Fname
FROM    EMPLOYEE
WHERE   Salary > ALL ( SELECT  Salary
                       FROM    EMPLOYEE
                       WHERE   Dno=5 );
```

Nested Queries (cont'd.)

- Avoid potential errors and ambiguities
 - Create tuple variables (aliases) for all tables referenced in SQL query

Query 16. Retrieve the name of each employee who has a dependent with the same first name and is the same sex as the employee.

```
Q16:  SELECT  E.Fname, E.Lname
      FROM    EMPLOYEE AS E
      WHERE   E.Ssn IN ( SELECT  Essn
                        FROM    DEPENDENT AS D
                        WHERE   E.Fname=D.Dependent_name
                        AND E.Sex=D.Sex );
```

Correlated Nested Queries

- **Queries that are nested using the = or IN comparison operator** can be collapsed into one single block: E.g., Q16 can be written as:
 - **Q16A:**

SELECT	E.Fname, E.Lname
FROM	EMPLOYEE AS E, DEPENDENT AS D
WHERE	E.Ssn=D.Essn AND E.Sex=D.Sex
	AND
	E.Fname=D.Dependent_name;
- **Correlated nested query**
 - Evaluated once for each tuple in the outer query